Hannah John
Software Student

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Project Log

Rental Search Web Application (React + API Integration)
- Built a full-stack rental property web app using React, Vite, JavaScript, and REST APIs.
- Implemented property search, filtering, authentication, ratings systems, and dynamic API data handling.
- Developed frontend components, management workflows, and reusable UI architecture.
- Worked with structured data models, asynchronous requests, responsive design, debugging, and user-focused interface design.

UAV Payload (Team Project)
- Developed the web dashboard and visualization interface for a UAV payload systems project.
- Connected frontend components to live API endpoints and realtime sensor data from Enviro+ board.
- Assisted with realtime graphing, telemetry monitoring, environmental sensor integration, and systems debugging.
- Worked within a collaborative Agile-style development environment using software practices.

Embedded Motor Control System (FreeRTOS - Team Project)
- Worked on a FreeRTOS-based embedded systems project using the Tiva TM4C1294 microcontroller.
- Developed motor control functionality, ADC filtering systems, and realtime telemetry features.
- Assisted with embedded software design, UART debugging, RPM monitoring, and hardware/software integration.
- Applied embedded-style programming, signal processing, and task management concepts.

Digital Voice Recorder Circuit Design
- Designed and evaluated analogue conditioning circuitry for a digital voice recorder system.
- Worked with op-amps, microphone input conditioning, ADC interfacing, and signal filtering.
- Focused on hardware implementation, testing, and performance evaluation/documentation.

Two-Toned Siren PCB Design
- Designed and implemented a two-toned siren circuit and PCB for an embedded electronics project.
- Worked on schematic design, component selection, PCB layout, and circuit testing.
- Assisted with prototyping, debugging, and evaluating circuit performance against design requirements.

Machine Learning Person Re-Identification System

- Built and evaluated machine learning models for person re-identification using PCA and Siamese CNN approaches.
- Implemented feature extraction, ranking systems, similarity matching, and Top-N retrieval evaluation methods.
- Worked with TensorFlow/Keras, image preprocessing pipelines, embedding generation, and model performance analysis.
- Applied analytical problem-solving, data processing, and machine learning workflow development.

Multi-Task Image Classification & Segmentation Project

- Developed CNN-based multitask learning models for pet breed classification and image segmentation.
- Trained both from-scratch DCNN and fine-tuned MobileNetV3 models using TensorFlow/Keras.
- Analysed model performance using IoU, Dice score, classification accuracy, and training evaluation metrics.

Personal Productivity Tracker (REST API Project)

- Building a cloud-based productivity tracking application using REST API architecture.
- Implementing analytics workflows, task tracking systems, automated reporting, and structured database management.
- Developing scalable backend logic, API endpoints, and data-driven productivity insights.
- Focused on software architecture, maintainability, and user-centered functionality.

Resume & Employer Matching Platform Concept

- Designed a startup-style platform focused on AI-assisted recruitment for students and graduates.
- Planned features including resume databases, employer dashboards, ranking systems, and AI matching.
- Worked on platform structure, database concepts, and user workflow planning.

Portfolio Website

- Built a personal portfolio website showcasing software engineering and full-stack development projects.
- Focused on responsive frontend design, project presentation, and clean UI/UX structure using Figma.
- Used HTML, CSS, JavaScript, and modern frontend tooling.

Art Portfolio Management Platform (Team Project)

- Contributed to the development of a collaborative art portfolio and management platform.
- Worked heavily on SQL database design, structured data handling, and backend integration.
- Assisted with frontend functionality, object-oriented application structure, and reusable component design.
- Participated in user testing, usability improvements, debugging, and overall system refinement.
- Collaborated within a team-based development environment using Git version control workflows.